

Operating System — Detailed Lecture Notes

JKSSB Computer Awareness

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19.1 What an operating system is

A computer's hardware cannot be used directly by a person or by ordinary application programs. Something must sit between the user and the machine, accept the user's instructions, and control the hardware on the user's behalf. That something is the **operating system**. An operating system (OS) is the main piece of **system software** that controls the whole computer. It starts the machine, runs other programs, and manages every resource. It makes the computer usable, because without an OS the hardware alone cannot run application software.

Definition (Operating system). *The main system software that sits between the user and the computer hardware, provides a user-friendly interface, and manages the computer's resources such as the processor, memory, storage, and input/output devices.*

Term	Meaning in this lesson
Kernel	The core part of the OS that interacts directly with the hardware.
Bootting	Starting the computer and loading the OS into main memory.
GUI	Graphical User Interface — interaction through icons, windows, and clicks.
CLI	Command Line Interface — interaction by typing text commands.
Process	A program that is currently executing.
Multitasking	Running more than one program at a time.
Time-sharing	Sharing one computer among several users at the same time.

The rest of this lesson uses these terms in one consistent sense. The **kernel** is the privileged core of the OS. **Bootting** is the startup sequence. A **GUI** is a graphical interface and a **CLI** is a text-command interface. A **process** is a running program. **Multitasking** and **time-sharing** are two ways of sharing the CPU.

19.2 The OS as interface and resource manager

The operating system has **two main jobs**. The first is to provide a **user-friendly interface**: the user gives instructions and receives results through the OS, without needing to understand the hardware. This is the definition of the OS as an interface layer, and it is a regularly tested idea (PYQ-21). The second job is to act as a **resource manager**: the OS shares the CPU, main memory, storage, and devices among all the programs that are running, so that each one gets a fair turn and none interferes with the others.

It is important to know **where** the operating system is. The OS is stored permanently on secondary storage, such as an SSD or HDD. At startup it is loaded into **primary storage**, that is,

main memory. While the computer is running, the OS is *resident in primary storage* (PYQ-22): it stays in main memory for the whole session so that it can respond instantly whenever the hardware or an application needs it.

Example 19.2.1. *An item states that the OS is kept in secondary storage while the computer is running. This is wrong. The OS is stored on secondary storage when the machine is off, but once the computer starts it is copied into primary storage (main memory) and remains there while running. The wrong option confuses the place where the OS is stored with the place where it runs.*

19.3 Functions of an operating system

The work of an OS is usually grouped into a few large functions, each of which is a favourite exam topic:

1. **Process management:** creating processes, scheduling the CPU among them, and removing them when they finish.
2. **Memory management:** deciding how much main memory each program gets and keeping programs apart.
3. **File management:** organising data into files and folders and handling their creation, reading, writing, and deletion.
4. **Device management:** controlling input and output devices through device drivers.
5. **Security:** managing user accounts, logins, and permissions, and applying updates.
6. **User interface:** presenting either a GUI or a CLI to the user.

19.4 The kernel

The **kernel** is the **core** of the operating system. It is the part of the OS that interacts **directly with the hardware**, and it always stays in main memory while the computer runs. The kernel provides the most important services: process scheduling, memory handling, and device control. Other software, including the user interface, relies on the kernel to reach the hardware.

Definition (Kernel). *The central, privileged part of the operating system that manages the hardware and provides the fundamental services on which all other software depends.*

19.5 User interfaces: GUI and CLI

Two broad kinds of interface are used to work with a computer. A **Graphical User Interface (GUI)** lets the user interact through pictures, icons, windows, menus, and clicks; it is easier for beginners. A **Command Line Interface (CLI)** requires the user to type text commands; it gives precise control but demands that the commands be known. Windows, macOS, and Linux desktops are examples of GUIs, while the command prompt and the terminal are examples of CLIs.

Point	GUI	CLI
Full form	Graphical User Interface	Command Line Interface
Input method	Clicks, icons, windows	Typed text commands
Ease of use	Easier for beginners	Needs the commands to be known
Example	Windows desktop, icons	Command prompt, terminal

19.6 Booting the computer

Booting is the process of starting a computer and loading the operating system into main memory. A **cold boot** starts from the power-off state, while a **warm boot** restarts a computer that is already on. The startup firmware, **BIOS** (Basic Input/Output System) or its modern replacement **UEFI**, first checks the hardware and then finds and loads the operating system from secondary storage. BIOS/UEFI initialises the hardware and hands control to the OS; it is not the kernel and it is not a device driver.

19.7 Process management

A **process** is a program that is currently executing. A program sitting on disk is passive; once it is loaded and running it becomes an active process. The operating system creates processes, gives each a turn on the CPU through **CPU scheduling**, lets them communicate where necessary, and removes them when they finish. Scheduling decides which process runs next and for how long, which is what allows several programs to make progress.

19.8 Memory management

The operating system decides **how much main memory** each program receives. It keeps one program's memory **separate** from another's so that programs cannot overwrite each other's data. It frees memory when a program ends so that the space can be reused. Where needed, the OS uses **virtual memory** to give programs more space than the physical RAM provides.

19.9 File management

The OS organises data on storage devices into **files** and **folders**. It handles creating, reading, writing, renaming, and deleting files, and it keeps a **directory** that records where each file is stored. It also controls **permissions**: which accounts may open, change, or delete a given file. Together these tasks are called the **file system**, the way the OS names, stores, and finds files.

19.10 Device management

The operating system controls all **input/output devices**: the keyboard, mouse, printer, disk, and so on. Each device communicates with the OS through a **device driver**. A device driver is **software**, not hardware — a point that is regularly tested (TRAP-10). The OS also uses **spooling** to queue print jobs and feed the printer at the printer's own speed, so that the computer does not have to wait for a slow device to finish.

19.11 Multitasking and time-sharing

Multitasking means that the operating system runs **more than one program at a time** (PYQ-23). The CPU switches between programs so quickly that they appear to run together, even though a single CPU core does only one thing at any instant. **Time-sharing** lets **many users** share one computer at the same time, each receiving a slice of CPU time in turn. Both multitasking and time-sharing depend on the OS **scheduler**.

Example 19.11.1. *An item defines multitasking as using more than one monitor, or as storing more than one file on a disk. Both are wrong. Multitasking is about the **processor** running more than one program at a time. The number of monitors and the number of files stored are unrelated to multitasking.*

19.12 Security, user accounts and updates

The operating system provides **user accounts**, each with a username and password. **Authentication** (the login step) verifies who is using the computer. **Permissions** decide which files and settings each account may use, so that one user cannot disturb another's work. Finally, system and software **updates** should be applied regularly, because they fix security holes and add improvements. Keeping the OS up to date is one of the simplest ways to protect the machine.

19.13 Examples of operating systems

Common operating systems include **Microsoft Windows**, the open-source **Linux** family, and Apple's **macOS** on desktops and laptops, plus the mobile operating systems **Android** and **iOS**. Android is built on the Linux kernel. An application such as MS Word or a web browser is *not* an operating system; it runs on top of one.

19.14 Exam retrieval and common confusions

The following distinctions prevent most errors on this topic.

1. The OS is **system software**, not application software. A device driver is **software**, not hardware (TRAP-10).
2. While running, the OS is resident in **primary storage** (main memory); it is only *stored* on secondary storage (PYQ-22).
3. The OS provides a **user-friendly interface** between the user and the hardware (PYQ-21).
4. **Multitasking** means running more than one program at a time; **time-sharing** shares one machine among several users (PYQ-23).
5. **BIOS/UEFI** initialises the hardware and loads the OS; it is not the kernel and not a device driver (TRAP-24).
6. A **program** on disk is passive; once running it becomes a **process**.

Example 19.14.1. *An item asks where the operating system is loaded when the computer starts. Trace the startup sequence: the machine is switched on, BIOS/UEFI checks the hardware, and the OS is loaded from secondary storage into **primary storage** (main memory), where it remains while the computer runs. If the stem instead asks which component is the core of the OS, the answer is the **kernel**.*

Exercises

1. Define an operating system and state its two main roles.
2. Where is the operating system stored, and where is it resident while the computer is running?
3. List four core functions of an operating system.
4. What is the kernel, and why is it important?
5. Distinguish a GUI from a CLI.
6. What does multitasking mean, and how does it differ from time-sharing?
7. State three ways an operating system helps to keep a computer secure.

Solutions

1. An operating system is the main system software that sits between the user and the hardware. Its two main roles are to provide a user-friendly interface and to manage the computer's resources.
2. It is stored permanently on secondary storage (such as an SSD or HDD) and is resident in primary storage (main memory) while the computer is running.
3. Any four of: process management, memory management, file management, device management, security, and the user interface.
4. The kernel is the core part of the operating system that interacts directly with the hardware. It is important because it provides the fundamental services — scheduling, memory handling, and device control — on which all other software depends.
5. A GUI (Graphical User Interface) uses icons, windows, and clicks and is easier for beginners; a CLI (Command Line Interface) requires the user to type text commands and gives precise control.
6. Multitasking means the OS runs more than one program at a time by switching the CPU among them. Time-sharing means many users share one computer at the same time, each getting a slice of CPU time.
7. Three of: user accounts with usernames and passwords; authentication at login; permissions that limit access to files and settings; and regular security updates.